AUTO TICKET CLASSIFICATION USING FLOW DESIGNER

PHASE 3

PROJECT DESIGN PHASE

1. Introduction

Auto Ticket Classification using FLOW DESIGNER is a project designed to automate the classification and routing of support tickets. In a typical support environment, tickets are created for issues related to applications, hardware, software, network access, accounts, and other services. Manually reviewing every ticket can consume time and may result in inconsistent categorization.

The proposed system uses a workflow-based approach in FLOW DESIGNER to capture ticket information, identify relevant classification attributes, assign a suitable category and priority, and route the ticket to the appropriate support group. The design emphasizes automation, consistency, traceability, and faster handling of incoming tickets.

2. Project Overview

The project provides a structured workflow for processing support tickets from creation to classification and assignment. The system receives ticket details such as short description, description, service, requester information, impact, urgency, and other available fields.

Based on predefined rules and classification logic, the workflow determines the ticket category, subcategory, priority, and assignment group. The classification result is then stored with the ticket and can be used for subsequent support activities.

3. Problem Statement

Organizations may receive a large number of support tickets every day. Manual classification requires support staff to read each ticket, understand the issue, select the correct category, and assign it to a suitable team. This process can introduce delays and human error.

The project addresses this problem by designing an automated FLOW DESIGNER workflow that standardizes ticket classification and reduces repetitive manual work.

4. Objectives

The major objectives of the project are:

- Automate the classification of incoming support tickets.
- Apply consistent classification rules to ticket information.
- Determine suitable category, priority, and assignment group.
- Reduce manual effort and ticket-routing delays.
- Maintain a clear and traceable workflow for ticket processing.
- Provide a design that can be extended with more advanced classification logic.

5. Proposed System

The proposed system consists of a ticket intake stage, data validation stage, classification stage, decision stage, assignment stage, and update/notification stage. The workflow is triggered when a new ticket is created or when a ticket enters the configured classification process.

The workflow checks required fields, evaluates ticket information against classification conditions, updates the ticket with the resulting values, and routes the ticket to the appropriate support group.

6. System Architecture

The system architecture is organized into the following logical layers:

6.1 Ticket Input Layer

Receives ticket information from the service-management environment.

6.2 Validation Layer

Checks whether the required ticket fields are available and usable.

6.3 Classification Layer

Evaluates ticket information using predefined conditions or classification rules.

6.4 Decision Layer

Determines category, subcategory, priority, and assignment group based on the classification outcome.

6.5 Action Layer

Updates the ticket and performs routing or notification actions.

6.6 Audit Layer

Preserves workflow execution information and ticket updates for traceability.

Figure 1. High-level system architecture

7. Workflow Design

The workflow can be represented by the following sequence:
1. Start workflow when a new ticket is available.
2. Read ticket details.
3. Validate required fields.
4. Normalize or inspect the ticket description and relevant attributes.
5. Apply classification conditions.
6. Determine category and subcategory.
7. Determine priority using impact and urgency or configured rules.
8. Select the appropriate assignment group.
9. Update the ticket with classification results.
10. Notify or route the ticket when required.
11. End workflow and retain execution information.

Figure 2. Ticket classification workflow

8. Module Design

8.1 Ticket Intake Module

Captures the incoming ticket and makes its fields available to the workflow.

8.2 Validation Module

Verifies that mandatory information is present. Missing information can be flagged for correction or sent through an exception path.

8.3 Classification Module

Maps ticket information to predefined categories. Example classification areas include hardware, software, network, access, application, and service requests.

8.4 Priority Module

Calculates or assigns priority using configured business rules involving impact, urgency, service criticality, or other approved criteria.

8.5 Assignment Module

Routes the ticket to a suitable support group according to the classification result.

8.6 Notification Module

Sends configured notifications or creates workflow actions after classification and assignment.

9. Data Flow

The primary data flow begins with the ticket record. Ticket attributes are passed into the workflow, validated, processed by classification rules, and converted into classification outputs. These outputs are written back to the ticket and used for routing.

Input Data: Ticket number, short description, description, requester, service, impact, urgency, and other configured fields.

Processing Data: Classification keywords, rule conditions, category mappings, priority rules, and assignment mappings.

Output Data: Category, subcategory, priority, assignment group, classification status, and workflow execution status.

Figure 3. Data flow of the proposed system

10. User Interface Design

The project can use the standard ticket interface available in the service-management environment. The design should make classification results easy to identify. Relevant fields may include category, subcategory, priority, assignment group, classification status, and any classification notes.

The interface should also provide clear status information so that support users can understand whether a ticket has been classified, requires manual review, or has entered an exception path.

11. Classification Logic

Classification logic is implemented using conditions and decision paths in FLOW DESIGNER. A rule can inspect one or more ticket attributes and produce a classification result.

Example: If a ticket contains information indicating a network connectivity issue, the workflow can classify it under Network and route it to the appropriate network support group. If the information is insufficient or does not match a configured rule, the workflow can send the ticket to a manual-review path.

Rules should be maintained in a structured and understandable form so that future team members can modify classifications without redesigning the complete workflow.

12. DATABASE / DATA DESIGN

The project can use the existing ticket data structure of the service-management platform. No unnecessary duplication of ticket information is required. Classification-related fields should be stored in the ticket record or an approved related data structure.

Important data elements include ticket identifier, description, category, subcategory, priority, assignment group, classification status, timestamps, and workflow execution details. Access to data should follow the permissions configured in the platform.

13. Testing Strategy

Testing should verify both normal and exception paths. Representative test tickets should be prepared for each major category and for combinations of impact and urgency where priority rules are used.

Test cases should verify:
• Correct classification for valid ticket descriptions.
• Correct routing to the expected assignment group.
• Correct priority based on configured rules.
• Handling of missing or incomplete information.
• Handling of tickets that do not match any rule.
• Correct ticket updates after workflow execution.
• Repeatability and consistency of classification results.

14. Expected Results

The expected result is a workflow that classifies and routes tickets in a consistent manner with reduced manual intervention. The system should improve the organization of incoming tickets, support faster assignment, and provide a clear record of the classification process.

15. Advantages and Limitations

Advantages

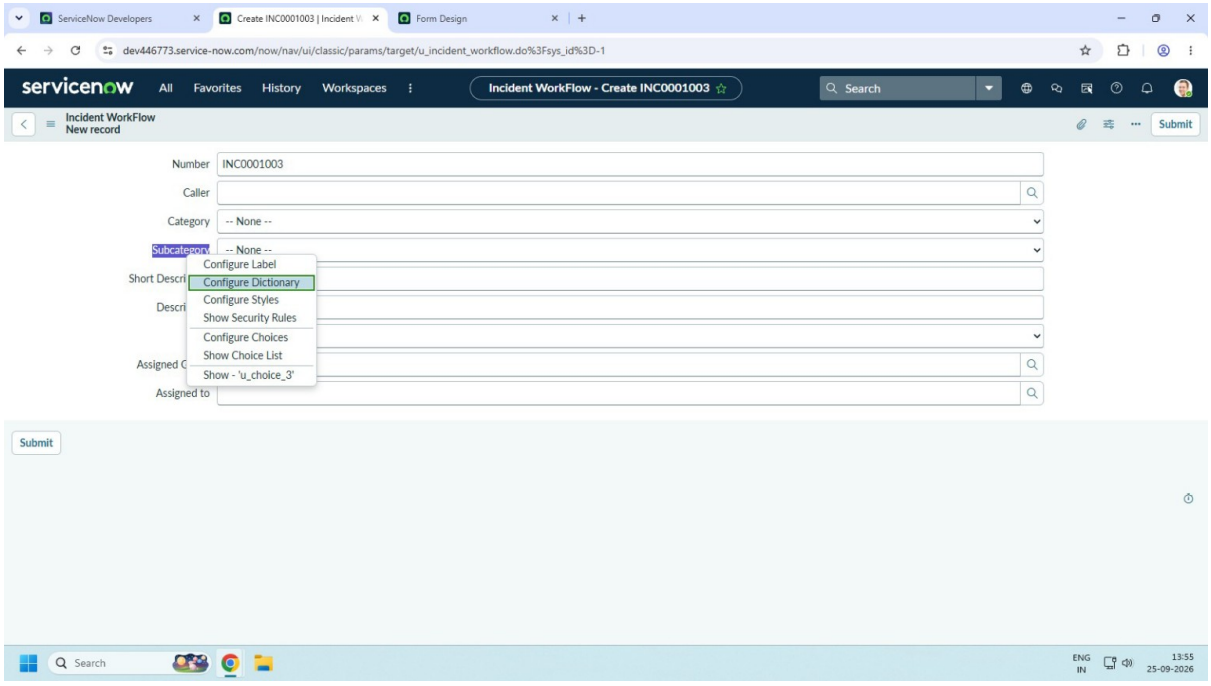
• Reduces repetitive manual classification work.
• Provides consistent application of configured rules.
• Supports faster ticket routing.
• Improves workflow traceability.
• Can be extended with additional categories and rules.

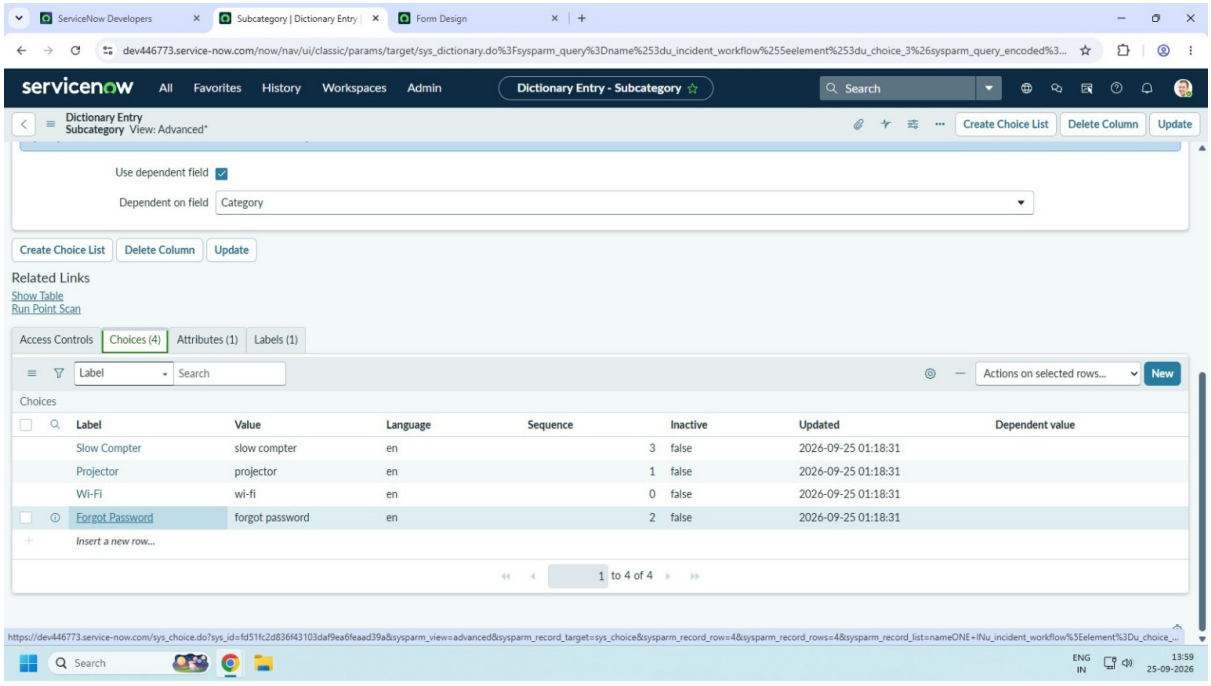
Limitations

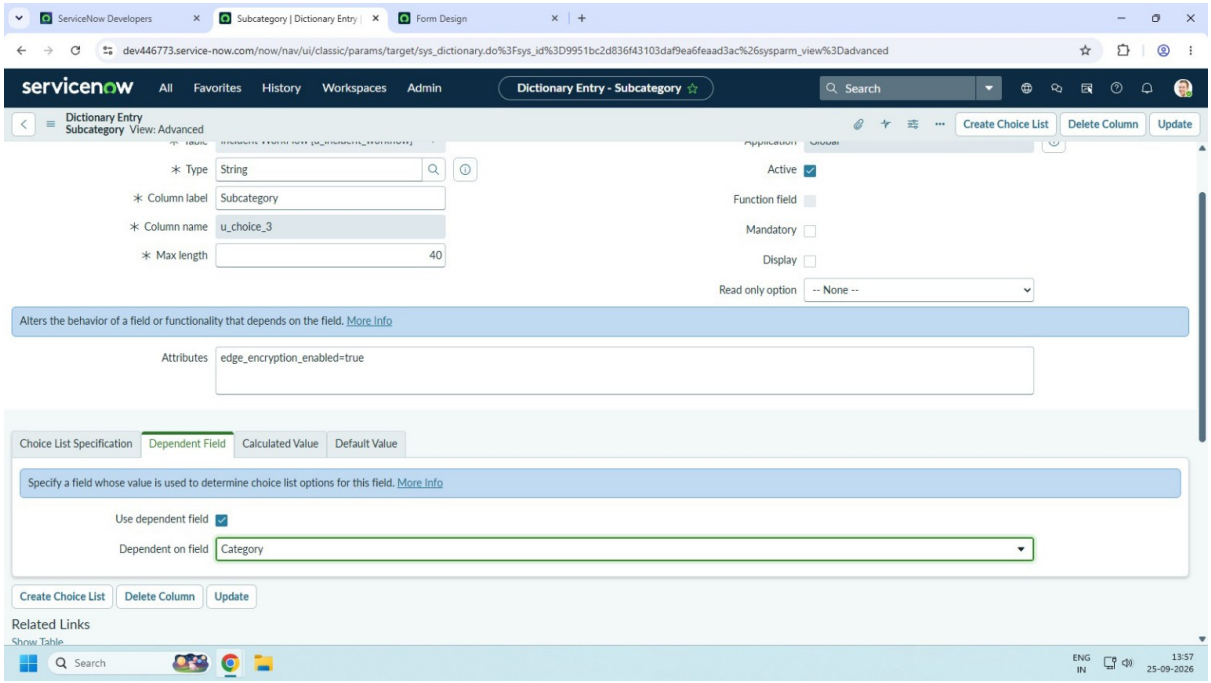
• Rule-based classification depends on the quality of configured conditions.
• Ambiguous or incomplete ticket descriptions may require manual review.
• Changes in service categories or organizational teams require rule maintenance.
• Advanced natural-language classification may require additional technologies or integrations.

16. Future Enhancements

Future versions may include machine-learning or natural-language classification, confidence scoring, multilingual ticket processing, analytics dashboards, automatic knowledge-article recommendations, duplicate-ticket detection, and feedback-based improvement of classification rules.


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Form Design
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servicenow
All
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Subcategory
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Show Choice List
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ServiceNow Developers
Subcategory | Dictionary Entry
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servicenow
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Workspaces
Admin
Dictionary Entry - Subcategory
Search
Dictionary Entry
Subcategory View: Advanced*
Create Choice List
Delete Column
Update
Use dependent field
Dependent on field
Category
Create Choice List
Delete Column
Update
Related Links
Show Table
Run Point Scan
Access Controls
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Attributes (1)
Labels (1)
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Dependent value
Slow Compter
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Subcategory | Dictionary Entry
Form Design
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History
Workspaces
Admin
Dictionary Entry - Subcategory
Search
Dictionary Entry
Subcategory View: Advanced
Create Choice List
Delete Column
Update
Type
String
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Column name
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Active
Function field
Mandatory
Display
Read only option
-- None --
Alters the behavior of a field or functionality that depends on the field. More Info
Attributes
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Choice List Specification
Dependent Field
Calculated Value
Default Value
Specify a field whose value is used to determine choice list options for this field. More Info
Use dependent field
Dependent on field
Category
Create Choice List
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17. Conclusion

Auto Ticket Classification using FLOW DESIGNER provides a structured approach for automating the classification and routing of support tickets. The Phase 3 project design defines the major workflow stages, modules, data flow, classification logic, testing approach, and expected results. The design can serve as a foundation for implementation and later enhancement with more advanced automation techniques.